

When Quantization Stops Paying

Thermal limits on sustained large language model inference at the edge, and what they mean for deciding where inference should run.

GROUP	PRIMARY RESEARCHER	CO-RESEARCHER	RESEARCH AREA
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01 Group and roles

This project is carried out by a group of two students, as required.

- **Faheem (23i-0728)** — Primary Researcher. Owns the measurement harness, the experimental protocol, and the Overleaf manuscript.
- **Irtaza Kazmi (23i-6001)** — Co-Researcher. Owns the literature review, the analysis and prediction code, and the evaluation against baselines.

Both members contribute to experiment design, writing, and the final defence. Work is split by experiment track rather than by "one codes, one writes", so each of us can explain any part of the study. A contribution log is kept alongside the repository, and the Git history records who did what.

02 Research area and title

Approved direction: Edge–Cloud Intelligence — intelligent placement of work across device, edge and cloud using latency, energy, cost and GPU availability.

Working title: *When Quantization Stops Paying: Thermal Limits on Sustained Edge LLM Inference*

Alternative under consideration: *Thermal Drift and the Erosion of Quantization Gains in Edge LLM Serving.*

03 Problem and motivation

Systems that decide *where* an AI model should run — on the user's own device or in the cloud — make that decision by measuring how fast the device is. That measurement is taken once, while the device is cool and idle.

The problem is that the number goes stale. A laptop GPU running continuously heats up, and to protect itself it quietly slows down. Nothing appears to be wrong: the device reports a normal temperature and keeps working. But it is doing far less work than the number the placement decision was based on.

Our own measurements show this is not a small effect. On an RTX 3050 laptop GPU running a 1.5-billion-parameter model, throughput fell from 116 to 66 tokens per second over fifteen minutes — a **43% loss** — while the reported temperature moved by only 4 °C.

A placement decision made on a cold device can be badly wrong within minutes, and the usual warning signal — temperature — does not reveal it.

This matters because running models on local devices is increasingly attractive for cost and privacy reasons, and because one of the main techniques used to make that practical — *quantization*, or shrinking a model so it needs less memory and arithmetic — is almost always evaluated in short benchmark runs on cool hardware.

04 Initial literature and research gap

Three bodies of work are close to this problem, and each stops short of it.

Edge measurement studies. Recent work benchmarks sustained inference on edge platforms and confirms that devices throttle heavily under continuous load [1]. However, it tests a single model at a single quantization level, is measurement-only, and does not address placement. Studies of quantized models on edge hardware evaluate energy, accuracy and latency [3], but not how those trade-offs change as a device heats over a long session.

Energy benchmarks. Large cross-hardware energy studies characterise many models and GPUs [2], but evaluate only full-precision weights and explicitly list quantized models as future work. They also measure steady-state behaviour without a time axis, so performance decay is invisible.

Thermal-aware scheduling. Schedulers that react to heat exist for data centres [4], where room-level cooling and power-capping controls are available. Consumer and edge hardware exposes neither, and these systems monitor temperature, which our measurements suggest is the wrong signal. On-device energy managers such as EnerInfer [7] tune a single phone-class device rather than deciding where work should run.

Meanwhile, placement and offloading systems [5] choose between edge and cloud using a fixed service rate measured in advance, and dynamic-precision methods [6] switch quantization level to meet accuracy or resource targets without considering heat.

THE GAP

No existing work measures how *quantization level* changes a device's ability to sustain performance under continuous load, or connects that behaviour to the decision of where inference should run. Quantization is evaluated for speed and accuracy, never for thermal sustainability.

05 Primary research question

How does quantization level affect the throughput a consumer GPU can sustain under continuous LLM inference, and can steady-state performance be predicted from a short measurement well enough to make correct edge–cloud placement decisions?

Two supporting questions follow from it:

- Why does the advantage of a more aggressively quantized model shrink as the device heats — what is the mechanism?
- How much error does a placement policy incur by assuming performance is constant?

06 Objectives

1. **Characterise** peak and sustained throughput, energy per token, and heating and cooling behaviour across model sizes, quantization levels and two different GPUs, using a controlled cold-start protocol.
2. **Explain** the mechanism behind the loss of quantization advantage, by separating the effect of reduced clock speed from the effect of heat itself.
3. **Build and validate** a predictor that estimates sustained throughput from a short cold-start measurement, and measure its error against full-length ground truth.
4. **Evaluate** a placement policy driven by that predictor against a policy that assumes constant performance, measuring decision accuracy and service-level violations.

07 Proposed methodology

The study is experimental, on real hardware, with a public and reproducible harness.

MEASUREMENT PROTOCOL

Every run follows the same three phases: cool the GPU to a defined cold start, apply fifteen minutes of continuous inference, then log fifteen minutes of idle recovery. Telemetry — temperature, power, clock speed and throttle status — is sampled at 20 Hz through NVIDIA's management library. Throughput is recorded per request.

Controlling the cold start matters: an early uncontrolled run began at 88 °C and measured only half the true degradation, because it never observed peak performance.

EXPERIMENTAL MATRIX

FACTOR	LEVELS
Model size	0.5B, 1.5B, 3B parameters
Quantization	Q4, Q8, FP16
Hardware	RTX 3050 laptop (4 GB), RTX 4070 laptop (8 GB)
Repetitions	5 per configuration

Ambient temperature is recorded for every run, and configurations that do not fit entirely in GPU memory are reported separately, since partial use of the CPU changes the behaviour being measured.

ISOLATING THE MECHANISM

To show that the loss of quantization advantage comes from reduced clock speed rather than heat itself, we repeat key comparisons with the GPU clock fixed at a low frequency on a cool device. If the effect reproduces without heat, the mechanism is confirmed.

PREDICTOR AND PLACEMENT POLICY

A curve is fitted to the first 60–90 seconds of each run and used to predict steady-state throughput, then checked against the full fifteen-minute result. The predictor drives a placement policy compared against two baselines: one assuming constant performance (what current systems do), and one reacting to temperature (what thermal-aware systems do).

08 Preliminary results

The harness is built and a first sweep is complete, which establishes feasibility. All figures below are from single runs on the RTX 3050 and will be repeated five times each.

CONFIGURATION	PEAK TOK/S	SUSTAINED TOK/S	LOSS	CLOCK AT STEADY STATE
0.5B · Q4	243.6	179.9	26.2%	1181 MHz
1.5B · Q4	115.8	66.2	42.9%	839 MHz
1.5B · Q8	95.4	66.8	30.0%	832 MHz

Three observations motivate the study.

- **The quantization advantage disappears.** The lighter Q4 model is 21% faster than Q8 at cold start. After fifteen minutes the two are statistically indistinguishable (66.2 against 66.8 tokens per second, well within one standard deviation across roughly 90 requests each).
- **Temperature hides the problem.** Temperature settles roughly nine times faster than throughput does, so the device looks stable for the entire period in which most of the performance is lost.
- **The effect is real, not a measurement artefact.** A smaller 0.5B model sustains 180 tokens per second rather than 66, confirming the ceiling scales with the work being done rather than being a fixed overhead in the serving software.

A likely mechanism is that aggressive quantization trades memory traffic for extra arithmetic to unpack compressed weights. Throttling removes arithmetic capacity, so the more compute-dependent option loses its advantage first. Objective 2 is designed to confirm or refute this.

09 Expected metrics and outcomes

METRIC	PURPOSE
Peak and sustained throughput	Size of the degradation, per configuration
Time to steady state	How quickly a fresh measurement becomes stale
Energy per token	Whether efficiency and speed degrade together
Prediction error (MAPE)	Accuracy of the predictor against full-length runs
Placement decision accuracy	Correct choices versus an ideal informed policy
Service-level violation rate	Practical cost of assuming constant performance

EXPECTED OUTCOMES

- An open dataset of sustained-performance measurements across model sizes, quantization levels and two GPUs.
- Evidence on whether short benchmarks overstate quantization gains, and by how much.
- A validated predictor of sustained throughput, released as usable code.
- A quantified case for when inference must move to the cloud.

10 Project infrastructure

OVERLEAF

One shared project created from the instructor's template, named to the required convention, with both students editing the same manuscript and the instructor added as Editor for the full duration.

```
[Grp12_23i-0728_23i-6001_thermal_limits_edge_llm_inference]
Instructor access: basharat.hussain.v@isb.nu.edu.pk (Editor)
```

PUBLIC GITHUB REPOSITORY

Public and active since day one, containing the measurement harness, the analysis code, and the raw telemetry from every run.

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https://github.com/MFaheemS/Cloud-Project
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It currently holds the cold-start harness, the analysis script, and raw 20 Hz telemetry for five completed runs. Both students have write access, and a contribution log records who did what. No credentials or

private data are published. The final line of the abstract will carry this link, as required.

11 Feasibility and risks

The main feasibility question — whether the effect is measurable on available hardware — is already answered: the harness works and the effect is large and repeatable.

RISK	MITIGATION
Only two GPUs available, limiting how far results generalise	Claims are scoped to consumer laptop GPUs; hardware is described precisely
Ambient temperature is not controlled	Recorded per run and reported; runs repeated across different conditions
Large models do not fit in 4 GB of memory	Those configurations are reported separately, not mixed with GPU-only results
Clock locking may be unavailable on laptop hardware	Mechanism can also be inferred from the natural range of clocks observed during throttling
Fast-moving research area	Repository is public and timestamped; writing runs alongside experiments rather than after them

12 References

[1] P. Tummalapalli, S. Arayakandy, R. Pal, and K. Kundan. LLM Inference at the Edge: Mobile, NPU, and GPU Performance Efficiency Trade-offs Under Sustained Load. *arXiv:2603.23640*, 2026.

[2] M. Fadel Argerich, J. Fürst, and M. Patiño-Martínez. Watt Counts: Energy-Aware Benchmark for Sustainable LLM Inference on Heterogeneous GPU Architectures. *arXiv:2604.09048*, 2026.

[3] E. J. Husom, A. Goknil, M. Astekin, L. K. Shar, A. Kåsen, S. Sen, B. A. Mithassel, and A. Soylu. Sustainable LLM Inference for Edge AI: Evaluating Quantized LLMs for Energy Efficiency, Output Accuracy, and Inference Latency. *ACM Transactions on Internet of Things*, 6(4), 2025. doi:10.1145/3767742.

[4] J. Stojkovic, C. Zhang, Í. Goiri, E. Choukse, H. Qiu, R. Fonseca, J. Torrellas, and R. Bianchini. TAPAS: Thermal- and Power-Aware Scheduling for LLM Inference in Cloud Platforms. In *Proc. ASPLOS ’25*, Rotterdam, Netherlands. ACM, 2025. doi:10.1145/3676641.3716025.

[5] Z. Yang, Y. Yang, C. Zhao, Q. Guo, W. He, and W. Ji. PerLLM: Personalized Inference Scheduling with Edge-Cloud Collaboration for Diverse LLM Services. *arXiv:2405.14636*, 2024.

[6] F. Liu, Z. Wang, J. Xia, J. Zhao, S. Zhao, J. Li, J. Liu, L. Jiang, and H. Guan. FlexQuant: A Flexible and Efficient Dynamic Precision Switching Framework for LLM Quantization. *arXiv:2506.12024*, 2025.

[7] B. Zou, N. Liu, B. Sun, M. Mascherin, D. Roy, Y. Liu, Y. Peng, N. Jia, and H. Chen. EnerInfer: Energy-Aware On-Device LLM Inference. *arXiv:2606.23001*, 2026.

